

Problem 3 (Bulgarian solitaire), Cell 3: a general upper bound

Problem 3 team

Official setting.

A partition of a positive integer n is a weakly decreasing sequence $\lambda = (\lambda_1, \dots, \lambda_s)$ of positive integers with $\lambda_1 + \dots + \lambda_s = n$. The shift $\mathcal{B}(\lambda)$ is the partition whose parts are the positive numbers among $\lambda_1 - 1, \dots, \lambda_s - 1$ together with one extra part equal to s . Call λ cyclic if $\mathcal{B}^i(\lambda) = \lambda$ for some $i \geq 1$, and let $d_{\mathcal{B}}(\lambda) = \min\{i \geq 0 : \mathcal{B}^i(\lambda) \text{ is cyclic}\}$, $D_{\mathcal{B}}(n) = \max\{d_{\mathcal{B}}(\lambda) : \lambda \text{ is a partition of } n\}$. Write $T_k = k(k+1)/2$ and $\delta_k = (k, k-1, \dots, 2, 1)$. Every $n \geq 1$ satisfies $T_{k-1} < n \leq T_k$ for exactly one k , which we call the rank of n .

Section 1 restates these definitions in an equivalent form, proves that $d_{\mathcal{B}}$ is well defined and that the rank exists and is unique (Lemma 1.2, where we also write $n = T_{k-1} + r$, $1 \leq r \leq k$), and fixes further notation.

Official statement of the cell.

Prove that for every $k \geq 4$ and every non-triangular n with $T_{k-1} < n < T_k$, $D_{\mathcal{B}}(n) \leq k^2 - 2k - 1$, and determine $D_{\mathcal{B}}(T_k - 1)$ exactly. Determine also, for that n , which partitions attain the maximum.

Status. Partial: value and both bounds fully proved for the stated range; the complete set of extremal partitions is given only by an exact dynamical criterion (closed form open).

- The bound and the value are proved in full. The case $k = 4$ uses an exhaustive computation over the 67 partitions of 7, 8, 9, with the code in Appendix A.
- The partitions attaining the maximum are determined *exactly* by the criterion of Theorem 9.1. We also give an explicit extremal family and a proved direct necessary condition.
- Qualification: this is a dynamical criterion, not a closed-form description of all extremal partitions; that description is open (Section 10). If a closed form is required, this sub-part is partial.

Answer and where it is proved.

- For $k \geq 4$ and $T_{k-1} < n < T_k$: $D_{\mathcal{B}}(n) \leq k^2 - 2k - 1$ (Theorem 7.1, [4, Thm. 4.4(1)]).
- $D_{\mathcal{B}}(T_k - 1) = k^2 - 2k - 1$ for every $k \geq 4$ (Theorem 8.2), attained by the explicit family $\lambda^{(k)} = (k-1, k-2, k-2, k-3, \dots, 2, 1, 1)$ ($\lambda_1 = k-1$, $\lambda_2 = k-2$, $\lambda_i = k+1-i$ for $3 \leq i \leq k$, $\lambda_{k+1} = 1$).

- Which partitions attain it. For $k = 4$: only $(3, 2, 2, 1, 1)$. For $k \geq 5$: exactly those $\lambda \vdash T_k - 1$ for which $\mathcal{B}^{k^2-2k-3}(\lambda)$ has at least $k + 1$ parts; equivalently, the part counts along the orbit contain the ramps $(k - 1, k, \dots, k, k + 1)$ at the positions $jk, \dots, jk + j + 1$, $j = 1, \dots, k - 3$ (Theorem 9.1). Every extremal partition ($k \geq 5$) satisfies the direct conditions $\lambda_1 \leq k + 1$ and $\#\{m : \lambda_m \geq i\} \geq k + 3 - 2i$ for all $i \geq 1$ (Proposition 10.1).

The proof of the bound reproduces, completed in detail, Griggs–Ho [4, Lemma 4.3, Thm. 4.4]. Their proof of the tail lemma is only sketched there; we give a different complete proof. Sections 1–5 are shared with Cell 2 (and, up to Section 4, with Cells 4 and 5), and are repeated so that this document is self-contained.

1 Setting and notation

Throughout, $T_m = m(m + 1)/2$ for integers $m \geq 0$, and $[P] \in \{0, 1\}$ denotes the truth value of a statement P . A *partition* λ of $n \geq 1$ (written $\lambda \vdash n$) is a finite non-increasing sequence of positive integers $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_\ell$ with sum n ; $\ell = \ell(\lambda)$ is its number of parts, and we put $\lambda_i = 0$ for $i > \ell(\lambda)$. Partitions are identified with multisets of positive integers.

Definition 1.1 (Bulgarian solitaire). For $\lambda \vdash n$ let $\mathcal{B}(\lambda)$ be the partition whose parts are $\ell(\lambda)$ together with the numbers $\lambda_i - 1$ for those $i \leq \ell(\lambda)$ with $\lambda_i \geq 2$. Since $\ell(\lambda) + \sum_i (\lambda_i - 1) = n$, we have $\mathcal{B}(\lambda) \vdash n$. Put $\mathcal{B}^0 = \text{id}$ and $\mathcal{B}^{i+1} = \mathcal{B} \circ \mathcal{B}^i$. A partition μ is *cyclic* if $\mathcal{B}^P(\mu) = \mu$ for some $P \geq 1$. For $\lambda \vdash n$, $d_{\mathcal{B}}(\lambda)$ is the least $i \geq 0$ such that $\mathcal{B}^i(\lambda)$ is cyclic, and $D_{\mathcal{B}}(n) = \max\{d_{\mathcal{B}}(\lambda) : \lambda \vdash n\}$. A partition $\lambda \vdash n$ with $d_{\mathcal{B}}(\lambda) = D_{\mathcal{B}}(n)$ is called *extremal*.

$d_{\mathcal{B}}(\lambda)$ is well defined: n has finitely many partitions, so $\mathcal{B}^a(\lambda) = \mathcal{B}^{a+P}(\lambda)$ for some $a \geq 0$, $P \geq 1$, and then $\mathcal{B}^a(\lambda)$ is cyclic. If μ is cyclic with $\mathcal{B}^P(\mu) = \mu$, then $\mathcal{B}^P(\mathcal{B}(\mu)) = \mathcal{B}(\mu)$, so $\mathcal{B}(\mu)$ is cyclic. Hence $\mathcal{B}^i(\lambda)$ is cyclic for every $i \geq d_{\mathcal{B}}(\lambda)$ and for no $i < d_{\mathcal{B}}(\lambda)$.

Lemma 1.2 (rank). *Every $n \geq 1$ can be written in exactly one way as $n = T_{k-1} + r$ with integers $k \geq 1$ and $1 \leq r \leq k$. We call k the rank of n ; n is triangular iff $r = k$ (then $n = T_k$).*

Proof. $T_0 = 0 < T_1 < T_2 < \dots$ is strictly increasing and unbounded, so there is exactly one $k \geq 1$ with $T_{k-1} < n \leq T_k$; then $r = n - T_{k-1}$ satisfies $1 \leq r \leq T_k - T_{k-1} = k$. Conversely $n = T_{k-1} + r$ with $1 \leq r \leq k$ forces $T_{k-1} < n \leq T_k$, i.e. this k . If $n = T_m$ ($m \geq 1$), then $T_{m-1} < n \leq T_m$, so $k = m$ and $r = T_m - T_{m-1} = m = k$; conversely $r = k$ gives $n = T_k$. $\square$

2 Diagrams, diagonals, rotation, and Brandt’s theorem

This section follows Griggs–Ho [4, §2] (the idea goes back to Etienne [3]); we make their “left shifting decreases the diagonal index” precise with an explicit potential Φ .

Let $\mathbb{P} = \mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$; the *cell* $(i, j) \in \mathbb{P}$ has *row* i and *column* j . The *diagram* of λ is

$$Y(\lambda) = \{(i, j) \in \mathbb{P} : 1 \leq j \leq \ell(\lambda), 1 \leq i \leq \lambda_j\},$$

so column j has λ_j cells (columns are parts, as in [4]). $Y(\lambda)$ determines λ , since $\lambda_j = \#\{i : (i, j) \in Y(\lambda)\}$. Row 1 of $Y(\lambda)$ is $\{(1, j) : j \leq \ell(\lambda)\}$ and column 1 has λ_1 cells; in particular

$$(1, w + 1) \in Y(\lambda) \implies (1, w) \in Y(\lambda) \quad (w \geq 1). \tag{1}$$

For a finite sequence $a = (a_1, \dots, a_m)$ of non-negative integers let $X_a = \{(i, j) : 1 \leq j \leq m, 1 \leq i \leq a_j\}$. If a is non-increasing, then $X_a = Y(\nu)$ where ν is the partition formed by the non-zero entries of a .

For $w \geq 1$ the w -th diagonal is $D_w = \{(i, j) \in \mathbb{P} : i + j = w + 1\}$; it has the w cells $d_w(j) = (w + 1 - j, j)$, $1 \leq j \leq w$ ($d_w(j)$ is the cell of D_w in column j). $\mathbb{P}$ is the disjoint union of the D_w . Put $\Delta_0 = \emptyset$ and $\Delta_m = D_1 \cup \dots \cup D_m$ for $m \geq 1$. Column j of Δ_m consists of the cells (i, j) with $i \leq m + 1 - j$, so $\Delta_m = Y(\delta_m)$ with $\delta_m = (m, m - 1, \dots, 1)$, and $|\Delta_m| = T_m$.

Define $\rho : \mathbb{P} \rightarrow \mathbb{P}$ by $\rho(i, j) = (i - 1, j + 1)$ if $i \geq 2$ and $\rho(1, j) = (j, 1)$.

Lemma 2.1. (a) $\rho(d_w(j)) = d_w(j + 1)$ for $1 \leq j < w$ and $\rho(d_w(w)) = d_w(1)$. Hence ρ maps each D_w bijectively onto itself, ρ is a bijection of $\mathbb{P}$, and for $m \geq 0$, $\rho^m(d_w(j)) = d_w(j')$ where $1 \leq j' \leq w$ and $j' \equiv j + m \pmod{w}$. In particular ρ^w is the identity on D_w . (b) ρ preserves the quantity $i + j$ of a cell.

Proof. (a) For $j < w$ the cell $d_w(j) = (w + 1 - j, j)$ has row $w + 1 - j \geq 2$, so $\rho(d_w(j)) = (w - j, j + 1) = d_w(j + 1)$; and $\rho(d_w(w)) = \rho(1, w) = (w, 1) = d_w(1)$. So ρ acts on D_w as the cyclic shift $j \mapsto j + 1$ of column indices modulo w ; the rest follows. (b) $(i - 1) + (j + 1) = i + j$ and $j + 1 = 1 + j$. $\square$

Lemma 2.2 (one move). Let $\lambda \vdash n$, $s = \ell(\lambda)$ and $a = (s, \lambda_1 - 1, \lambda_2 - 1, \dots, \lambda_s - 1)$. Then $\rho(Y(\lambda)) = X_a$, and $Y(\mathcal{B}(\lambda)) = X_{a^*}$, where a^* is the non-increasing rearrangement of a . If $s \geq \lambda_1 - 1$, then a is non-increasing and $Y(\mathcal{B}(\lambda)) = \rho(Y(\lambda))$.

Proof. The cells of row 1 of $Y(\lambda)$ are $(1, j)$, $1 \leq j \leq s$; ρ sends them to $(j, 1)$, i.e. exactly onto the cells $(1, 1), \dots, (s, 1)$ of column 1. A cell $(i, j) \in Y(\lambda)$ with $i \geq 2$ satisfies $2 \leq i \leq \lambda_j$ and is sent to $(i - 1, j + 1)$; for fixed j these images are exactly the cells $(1, j + 1), \dots, (\lambda_j - 1, j + 1)$. Hence $\rho(Y(\lambda)) = X_a$. The parts of $\mathcal{B}(\lambda)$ are the non-zero entries of a , so $Y(\mathcal{B}(\lambda)) = X_{a^*}$. The entries of a after the first are non-increasing because λ is; so a is non-increasing iff $s \geq \lambda_1 - 1$, and then $a^* = a$. $\square$

For a finite set $X \subseteq \mathbb{P}$ put $\Phi(X) = \sum_{(i,j) \in X} (i + j)$.

Lemma 2.3. For every finite sequence a of non-negative integers, $\Phi(X_{a^*}) \leq \Phi(X_a)$, with equality iff a is non-increasing.

Proof. $\Phi(X_a) = \sum_j \sum_{i=1}^{a_j} (i + j) = \sum_j \binom{a_j + 1}{2} + F(a)$ with $F(a) = \sum_j j a_j$. The first sum does not change when a is permuted. If $a_j < a_{j+1}$, exchanging these two entries changes F by $j a_{j+1} + (j + 1) a_j - j a_j - (j + 1) a_{j+1} = a_j - a_{j+1} < 0$. Starting from a , repeatedly exchange some adjacent pair with $a_j < a_{j+1}$. Each exchange lowers the non-negative integer F , so the process stops, and it stops at a sequence with no adjacent ascent, i.e. at the non-increasing rearrangement a^* (which is unique). If a is not non-increasing at least one exchange happens, so $F(a^*) < F(a)$; if a is non-increasing then $a = a^*$. $\square$

Corollary 2.4. For $\lambda \vdash n$: $\Phi(Y(\mathcal{B}(\lambda))) \leq \Phi(Y(\lambda))$, with equality iff $\ell(\lambda) \geq \lambda_1 - 1$; in the case of equality $Y(\mathcal{B}(\lambda)) = \rho(Y(\lambda))$.

Proof. By Lemma 2.1, ρ is injective and preserves $i + j$, so $\Phi(\rho(X)) = \Phi(X)$. With a as in Lemma 2.2: $\Phi(Y(\lambda)) = \Phi(X_a) \geq \Phi(X_{a^*}) = \Phi(Y(\mathcal{B}(\lambda)))$, with equality iff a is non-increasing (Lemma 2.3), i.e. iff $\ell(\lambda) \geq \lambda_1 - 1$, and then Lemma 2.2 gives $Y(\mathcal{B}(\lambda)) = \rho(Y(\lambda))$. $\square$

For $k \geq 1$ and $E \subseteq D_k$ let λ_E be the partition whose j -th part, $1 \leq j \leq k$, is $k - j + [d_k(j) \in E]$ (zero parts omitted). The sequence $(k - j + \varepsilon_j)_{j=1}^k$ with $\varepsilon_j \in \{0, 1\}$ is non-increasing (consecutive

differences are $1 + \varepsilon_j - \varepsilon_{j+1} \geq 0$), so λ_E is a partition, $|\lambda_E| = T_{k-1} + |E|$, and $Y(\lambda_E) = \Delta_{k-1} \cup E$ (column j consists of the $k - j$ cells of Δ_{k-1} and, if $d_k(j) \in E$, the cell $d_k(j) = (k + 1 - j, j)$). Writing $e_{k-j} = [d_k(j) \in E]$ we get $\lambda_E = (k - 1 + e_{k-1}, k - 2 + e_{k-2}, \dots, 1 + e_1, e_0)$.

Theorem 2.5 (Brandt [2]; see also [1, 3, 4]). *Let $n = T_{k-1} + r$ with $1 \leq r \leq k$. A partition of n is cyclic if and only if it equals λ_E for some $E \subseteq D_k$ with $|E| = r$; equivalently, iff it has the form $(k - 1 + e_{k-1}, \dots, 1 + e_1, e_0)$ with $e_i \in \{0, 1\}$ and $\sum_i e_i = r$ (zero parts omitted). Moreover $Y(\mathcal{B}(\lambda_E)) = \rho(Y(\lambda_E))$, i.e. $\mathcal{B}(\lambda_E) = \lambda_{\rho(E)}$, and $\mathcal{B}^k(\lambda_E) = \lambda_E$.*

Proof. ($\Leftarrow$ and “moreover”). Let $\mu = \lambda_E$ with $E \subseteq D_k$, $|E| = r$. Row 1 of $Y(\mu) = \Delta_{k-1} \cup E$ consists of the cells $(1, w) = d_w(w)$ with $w \leq k - 1$ and possibly $d_k(k)$; so $\ell(\mu) = k - 1 + [d_k(k) \in E] \geq k - 1$. Column 1 consists of $d_w(1)$ for $w \leq k - 1$ and possibly $d_k(1)$; so $\mu_1 = k - 1 + [d_k(1) \in E] \leq k$. Thus $\ell(\mu) \geq \mu_1 - 1$ and Corollary 2.4 and Lemma 2.1 give $Y(\mathcal{B}(\mu)) = \rho(\Delta_{k-1} \cup \rho(E)) = \Delta_{k-1} \cup \rho(E)$ with $\rho(E) \subseteq D_k$, $|\rho(E)| = r$. Hence $\mathcal{B}(\lambda_E) = \lambda_{\rho(E)}$, and by induction $\mathcal{B}^k(\lambda_E) = \lambda_{\rho^k(E)} = \lambda_E$ because ρ^k is the identity on D_k . So μ is cyclic.

($\Rightarrow$). Let $\lambda \vdash n$ with $\mathcal{B}^P(\lambda) = \lambda$, $P \geq 1$, and write $\lambda^{(m)} = \mathcal{B}^m(\lambda)$. By Corollary 2.4, $\Phi(Y(\lambda^{(0)})) \geq \Phi(Y(\lambda^{(1)})) \geq \dots \geq \Phi(Y(\lambda^{(P)})) = \Phi(Y(\lambda^{(0)}))$, so all these are equalities and $Y(\lambda^{(m+1)}) = \rho(Y(\lambda^{(m)}))$ for $0 \leq m < P$. As $\lambda^{(m+P)} = \lambda^{(m)}$, this holds for all $m \geq 0$, and therefore $Y(\lambda^{(m)}) = \rho^m(Y(\lambda))$ for all $m \geq 0$.

Claim: there is no $w \geq 1$ such that $D_w \not\subseteq Y(\lambda)$ and $D_{w+1} \cap Y(\lambda) \neq \emptyset$. Suppose $d_w(\alpha) \notin Y(\lambda)$ and $d_{w+1}(\beta) \in Y(\lambda)$. Since $\gcd(w, w+1) = 1$, the Chinese remainder theorem gives $m \geq 0$ with $m \equiv w - \alpha \pmod{w}$ and $m \equiv w + 1 - \beta \pmod{w+1}$. By Lemma 2.1, $\rho^m(d_w(\alpha)) = d_w(w) = (1, w)$ and $\rho^m(d_{w+1}(\beta)) = d_{w+1}(w+1) = (1, w+1)$. As ρ^m is injective, $(1, w) \notin \rho^m(Y(\lambda)) = Y(\lambda^{(m)})$ while $(1, w+1) \in Y(\lambda^{(m)})$, contradicting (1).

$Y(\lambda)$ is finite, so some D_w is not contained in it; let $K \geq 1$ be the least such w . Then $\Delta_{K-1} \subseteq Y(\lambda)$. By the claim, $D_{K+1} \cap Y(\lambda) = \emptyset$; then $D_{K+1} \not\subseteq Y(\lambda)$, so again by the claim $D_{K+2} \cap Y(\lambda) = \emptyset$, and inductively $D_w \cap Y(\lambda) = \emptyset$ for all $w > K$. Thus $Y(\lambda) = \Delta_{K-1} \cup E'$ with $E' = Y(\lambda) \cap D_K \neq D_K$; let $r' = |E'| \in \{0, \dots, K - 1\}$, so $n = T_{K-1} + r'$ and $Y(\lambda) = Y(\lambda_{E'})$ (with k replaced by K in the definition of $\lambda_{E'}$). If $r' \geq 1$, then $T_{K-1} < n < T_K$, so by Lemma 1.2 $K = k$, $r' = r$, and $\lambda = \lambda_{E'}$. If $r' = 0$, then $K \geq 2$ (as $n \geq 1$) and $n = T_{K-1} = T_{K-2} + (K - 1)$, so $k = K - 1$, $r = k$, and $Y(\lambda) = \Delta_k = \Delta_{k-1} \cup D_k$, i.e. $\lambda = \lambda_{D_k}$ with $|D_k| = k = r$. $\square$

Corollary 2.6. *Let $n = T_{k-1} + r$, $1 \leq r \leq k$.*

- (a) *If $r = k$ (i.e. $n = T_k$), the only cyclic partition of n is $\delta_k = (k, k - 1, \dots, 1)$, and $\mathcal{B}(\delta_k) = \delta_k$.*
- (b) *Every cyclic $\mu \vdash n$ satisfies $Y(\mu) \subseteq \Delta_k$, $\Delta_{k-1} \subseteq Y(\mu)$, $\ell(\mu) \in \{k - 1, k\}$ and $\mathcal{B}^k(\mu) = \mu$.*
- (c) *If $\mu \vdash n$ is cyclic, then for every $a \geq 0$: $\sum_{i=a}^{a+k-1} \ell(\mathcal{B}^i(\mu)) = k(k - 1) + r$. If moreover $r < k$, both values $k - 1$ and k occur among $\ell(\mathcal{B}^a(\mu)), \dots, \ell(\mathcal{B}^{a+k-1}(\mu))$.*

Proof. (a) and (b) are immediate from Theorem 2.5 ($E = D_k$ when $r = k$; $\ell(\lambda_E) = k - 1 + [d_k(k) \in E]$ was computed in its proof). (c) Write $\mu = \lambda_E$. Then $\mathcal{B}^i(\mu) = \lambda_{\rho^i(E)}$, so $\ell(\mathcal{B}^i(\mu)) = k - 1 + [d_k(k) \in \rho^i(E)] = k - 1 + [d_k(j_i) \in E]$, where $d_k(j_i) = \rho^{-i}(d_k(k))$, i.e. $j_i \equiv k - i \pmod{k}$ (Lemma 2.1). As i runs through $a, \dots, a + k - 1$, j_i runs through all of $1, \dots, k$ once, so the sum is $k(k - 1) + |E| = k(k - 1) + r$. If $0 < r < k$, some j_i lies in E and some does not. $\square$

3 Piles and the part-count sequence

This is the “ $\text{diagram}_B(\lambda)$ ” bookkeeping of Griggs–Ho [4, §3], written with explicitly labelled piles.

Fix $\lambda \vdash n$, write $\lambda^{(i)} = \mathcal{B}^i(\lambda)$ and

$$c_i = \ell(\lambda^{(i-1)}) \quad (i \geq 1).$$

We call $c(\lambda) = (c_1, c_2, \dots)$ the *part-count sequence*; clearly $c(\lambda^{(a)}) = (c_{a+1}, c_{a+2}, \dots)$. Every $c_i \geq 1$.

Lemma 3.1 (piles). *For every $i \geq 0$ the multiset of parts of $\lambda^{(i)}$ is*

$$\{\lambda_m - i : 1 \leq m \leq \ell(\lambda), \lambda_m > i\} \uplus \{c_j - (i - j) : 1 \leq j \leq i, c_j > i - j\}.$$

Proof. Induction on i ; $i = 0$ is clear. $\lambda^{(i+1)} = \mathcal{B}(\lambda^{(i)})$ has the part $\ell(\lambda^{(i)}) = c_{i+1} = c_{i+1} - 0$ and the parts of $\lambda^{(i)}$ decreased by one, the zeros being discarded: this is the stated multiset for $i + 1$ (the term $j = i + 1$ is c_{i+1}). $\square$

We use *labels*: $O_1, \dots, O_{\ell(\lambda)}$ (the original piles) and $N_1, N_2, \dots$ (N_j is the pile created by the j -th move, of size c_j when created). By Lemma 3.1, at time i (i.e. in $\lambda^{(i)}$) the pile O_m has size $\lambda_m - i$ and the pile N_j ($j \leq i$) has size $c_j - (i - j)$, a pile being present iff its size is positive. For $i \geq 1$ let R_i be the set of labels of the piles present at time $i - 1$:

$$O_m \in R_i \iff \lambda_m \geq i, \quad N_j \in R_i \iff j \leq i - 1 \text{ and } c_j \geq i - j. \quad (2)$$

By Lemma 3.1 (with $i - 1$), $|R_i| = \ell(\lambda^{(i-1)}) = c_i$. Let $Z_i = R_i \setminus R_{i+1}$ (the piles of size 1 at time $i - 1$, which disappear in move i).

Lemma 3.2 (basic facts). *For all $i \geq 1$:*

- (a) $N_i \in R_{i+1}$ and $N_i \notin R_i$.
- (b) $R_{i+1} = (R_i \setminus Z_i) \cup \{N_i\}$, hence $c_{i+1} = c_i + 1 - |Z_i|$ and $c_{i+1} \leq c_i + 1$.
- (c) If $c_{i+1} = c_i + 1$ then $Z_i = \emptyset$, i.e. $R_i \subseteq R_{i+1}$. If $c_{i+1} = c_i$ then $|Z_i| = 1$.
- (d) For $j < i$: $N_j \in R_i$ implies $N_j \in R_{i'}$ for $j < i' \leq i$, and $N_j \notin R_i$ implies $N_j \notin R_{i'}$ for all $i' \geq i$. Likewise for the labels O_m .

Proof. (a) $c_i \geq 1 = (i + 1) - i$; and labels N_j in R_i have $j \leq i - 1$. (d) For $i' > j$ the clause $j \leq i' - 1$ in (2) holds automatically, so membership of N_j in $R_{i'}$ is the condition $c_j \geq i' - j$, which is monotone in i' ; for O_m the condition $\lambda_m \geq i'$ is monotone as well. (b) By (d), $R_{i+1} \setminus \{N_i\} \subseteq R_i$, so $R_{i+1} = (R_i \cap R_{i+1}) \cup \{N_i\} = (R_i \setminus Z_i) \cup \{N_i\}$, a disjoint union by (a); take cardinalities. (c) follows from $c_{i+1} = c_i + 1 - |Z_i|$. $\square$

Lemma 3.3 (gap lemma). *Let $1 \leq j < i$. If $N_j \notin R_i$ and $N_{j+1} \in R_{i+1}$, then $c_j = i - j - 1$, $c_{j+1} = i - j$, and $N_{j+1} \notin R_{i+2}$.*

Proof. By (2), $c_j \leq i - j - 1$ and $c_{j+1} \geq (i + 1) - (j + 1) = i - j$. By Lemma 3.2(b), $c_{j+1} \leq c_j + 1 \leq i - j$. Hence both equalities hold, and $c_{j+1} = i - j < (i + 2) - (j + 1)$ gives $N_{j+1} \notin R_{i+2}$. $\square$

Lemma 3.4 (the sequence determines the partition). *For $i \geq 1$, $\#\{m : \lambda_m \geq i\} = c_i - \#\{j : 1 \leq j \leq i - 1, c_j \geq i - j\}$. Consequently, if two partitions have the same part-count sequence, they are equal.*

Proof. The formula is $|R_i| = c_i$ together with (2). The numbers $\#\{m : \lambda_m \geq i\}$, $i \geq 1$, determine λ , since $\lambda_m = \#\{i \geq 1 : \#\{m' : \lambda_{m'} \geq i\} \geq m\}$. $\square$

Lemma 3.5 (window sums; generalising [4, Prop. 3.2(2)]). *For all $a \geq 0$ and $h \geq 1$: $c_{a+1} + c_{a+2} + \dots + c_{a+h} \leq n + T_{h-1}$.*

Proof. Let $\mu = \lambda^{(a)}$; its part-count sequence is $(c_{a+1}, c_{a+2}, \dots)$. Applying (2) to μ , for $1 \leq g \leq h$ the number c_{a+g} equals $\#\{m : \mu_m \geq g\}$ plus the number of piles created in the first $g-1$ moves starting from μ that are still present, which is at most $g-1$. Summing, $\sum_{g=1}^h c_{a+g} \leq \sum_{g \geq 1} \#\{m : \mu_m \geq g\} + \sum_{g=1}^h (g-1) = |\mu| + T_{h-1} = n + T_{h-1}$. $\square$

4 Ramps

The following lemmas are Lemmas 3.4–3.6 and Proposition 3.2(3) of Griggs–Ho [4]; we give complete proofs (the proof of [4, Lemma 3.5] ends with “continue this process”, which we replace by an explicit induction) and we add the equality statement in Proposition 4.5.

Definition 4.1. A *ramp* of λ is a triple (p, q, x) of integers with $p \geq 1$, $q \geq p+2$ and

$$c_p = x - 1, \quad c_{p+1} = c_{p+2} = \dots = c_{q-1} = x, \quad c_q = x + 1.$$

Its *length* is $q - p$ and its *value* is x (note $x \geq 2$, since $c_p \geq 1$).

Lemma 4.2 (sandwich). *If $1 \leq i < j$ and $c_i < x < c_j$ for an integer x , then there is a ramp (p, q, x) with $i \leq p < q \leq j$.*

Proof. Let p be the largest m with $i \leq m < j$ and $c_m \leq x - 1$ (it exists: $m = i$ qualifies). By Lemma 3.2(b), $c_{p+1} \leq c_p + 1 \leq x$. Since $c_j > x$, this rules out $p + 1 = j$; so $p + 1 < j$, and $c_{p+1} \geq x$ by maximality of p . Hence $c_p = x - 1$ and $c_{p+1} = x$. Let q be the least m with $p < m \leq j$ and $c_m \geq x + 1$ (it exists: $m = j$ qualifies). Then $q \geq p + 2$ since $c_{p+1} = x$. For $p < m < q$ we have $c_m \leq x$ (minimality of q) and $c_m \geq x$ (maximality of p , as $m < j$), so $c_m = x$. Finally $x + 1 \leq c_q \leq c_{q-1} + 1 = x + 1$. $\square$

Lemma 4.3 (short ramps [4, Lemma 3.6]). *If $(p, p+2, x)$ is a ramp, then $p \leq x$.*

Proof. Suppose $p \geq x + 1$; then $p \geq 3$. Since $|R_p| = c_p = x - 1 \leq p - 2$, not all of $N_1, \dots, N_{p-1}$ lie in R_p , while $N_{p-1} \in R_p$ (Lemma 3.2(a)). Let j be the largest index in $[1, p-2]$ with $N_j \notin R_p$; then $N_{j+1} \in R_p$. As $c_{p+1} = c_p + 1$ and $c_{p+2} = c_{p+1} + 1$, Lemma 3.2(c) gives $R_p \subseteq R_{p+1} \subseteq R_{p+2}$, so $N_{j+1} \in R_{p+1}$ and $N_{j+1} \in R_{p+2}$. But Lemma 3.3 (with $i = p$) gives $N_{j+1} \notin R_{p+2}$, a contradiction. $\square$

Lemma 4.4 (descent [4, Lemma 3.5]). *Let (p, q, x) be a ramp with $q - p \geq 3$ and $p \geq x + 1$. Then there is a ramp (p', q', x') with*

$$p - x \leq p' \leq p - 2, \quad x' = p - p' \leq x, \quad 2 \leq q' - p' \leq q - p - 1.$$

Proof. The x labels $N_{p-x}, \dots, N_{p-1}$ exist since $p - x \geq 1$. As $|R_p| = x - 1$, one of them is not in R_p ; $N_{p-1} \in R_p$. Let p' be the largest index in $[p - x, p - 2]$ with $N_{p'} \notin R_p$ and put $y = p - p'$, so $2 \leq y \leq x$ and $N_{p'+1}, \dots, N_{p-1} \in R_p$. As $c_{p+1} = c_p + 1$, $R_p \subseteq R_{p+1}$, and with Lemma 3.2(a), $N_{p'+1}, \dots, N_p \in R_{p+1}$. Lemma 3.3 ($j = p'$, $i = p$) gives $c_{p'} = y - 1$ and $c_{p'+1} = y$.

For $1 \leq l \leq q - p - 1$ let $H(l)$ be the statement

$$H(l): \quad c_{p'+1} = \dots = c_{p'+l} = y \quad \text{and} \quad N_{p'+l}, N_{p'+l+1}, \dots, N_{p+l-1} \in R_{p+l}.$$

We just proved $H(1)$. Assume $H(l)$. Since $c_{p'+l} = y = p - p'$, (2) gives, for $i > p' + l$: $N_{p'+l} \in R_i \iff i \leq p + l$. Hence $N_{p'+l} \in Z_{p+l}$.

- If $l = q - p - 1$, then $p + l = q - 1$ and $c_q = c_{q-1} + 1$, so $Z_{q-1} = \emptyset$ by Lemma 3.2(c): a contradiction. So $H(q - p - 1)$ is false.
- If $l \leq q - p - 2$, then $p + l$ and $p + l + 1$ lie in $[p + 1, q - 1]$, so $c_{p+l+1} = c_{p+l} = x$ and $|Z_{p+l}| = 1$, i.e. $Z_{p+l} = \{N_{p'+l}\}$. By Lemma 3.2(b), $R_{p+l+1} = (R_{p+l} \setminus \{N_{p'+l}\}) \cup \{N_{p+l}\} \supseteq \{N_{p'+l+1}, \dots, N_{p+l}\}$. In particular $N_{p'+l+1} \in R_{p+l+1}$, i.e. $c_{p'+l+1} \geq (p+l+1) - (p'+l+1) = y$, and $c_{p'+l+1} \leq c_{p+l} + 1 = y + 1$. So either $c_{p'+l+1} = y + 1$, or $c_{p'+l+1} = y$ and then $H(l + 1)$ holds.

Let l_0 be the largest $l \leq q - p - 1$ such that $H(1), \dots, H(l)$ all hold; then $1 \leq l_0 \leq q - p - 2$, and since $H(l_0 + 1)$ fails, the second item gives $c_{p'+l_0+1} = y + 1$. Thus $(p', p' + l_0 + 1, y)$ is a ramp of length $l_0 + 1 \in [2, q - p - 1]$, and $x' = y$ has the stated properties. $\square$

Proposition 4.5 (ramp bound). *Every ramp (p, q, x) satisfies $p \leq (q - p - 1)x$. If equality holds, then $(jx, jx + j + 1, x)$ is a ramp for every $j = 1, 2, \dots, q - p - 1$.*

Proof. Strong induction on the length $L = q - p \geq 2$. If $L = 2$, Lemma 4.3 gives $p \leq x$; if $p = x$, the ramp itself is $(x, x + 2, x)$. Let $L \geq 3$. If $p \leq x$ then $p \leq x < (L - 1)x$. Otherwise Lemma 4.4 provides a ramp (p', q', x') with $p' \geq p - x$, $x' \leq x$ and length $L' \leq L - 1$; by induction $p' \leq (L' - 1)x'$, hence

$$p \leq p' + x \leq (L' - 1)x' + x \leq (L - 2)x + x = (L - 1)x.$$

If $p = (L - 1)x$, all these inequalities are equalities: $p' = p - x = (L - 2)x$ and $(L' - 1)x' = (L - 2)x$; since $1 \leq L' - 1 \leq L - 2$ and $x' \leq x$, this forces $L' = L - 1$ and $x' = x$. So $((L - 2)x, (L - 2)x + L - 1, x)$ is a ramp attaining equality, and by induction $(jx, jx + j + 1, x)$ is a ramp for $j = 1, \dots, L - 2$; for $j = L - 1$ it is (p, q, x) itself. $\square$

Lemma 4.6 (long ramps [4, Lemma 3.4]). *If (p, q, x) is a ramp with $q - p = x + 1$, then $q \leq n + 1$. (Recall that $x \geq 2$ for every ramp.)*

Proof. For $p + 1 \leq j \leq q - 1$ we have $c_j = x \geq q - j$, so $N_j \in R_q$: these are x piles. Also $c_p = x - 1 < q - p$, so $N_p \notin R_q$. Since $|R_q| = c_q = x + 1$, R_q contains exactly one further label P , and $P = O_m$ for some m , or $P = N_j$ with $1 \leq j \leq p - 1$. If $P = O_m$, then $q \leq \lambda_m \leq n$. If $P = N_1$, then $c_1 \geq q - 1$, and $c_1 = \ell(\lambda) \leq n$. Suppose $P = N_j$ with $2 \leq j \leq p - 1$. Then $N_{j-1} \notin R_q$. As $c_q = c_{q-1} + 1$, $R_{q-1} \subseteq R_q$, so $N_{j-1} \notin R_{q-1}$. Next, $q - 2 \geq p + 1$ (as $x \geq 2$), so $c_{q-1} = c_{q-2} = x$ and $|Z_{q-2}| = 1$; since $c_p = x - 1 = (q - 2) - p$, $N_p \in R_{q-2} \setminus R_{q-1}$, so $Z_{q-2} = \{N_p\}$. As $j - 1 \neq p$, $N_{j-1} \in R_{q-2}$ would give $N_{j-1} \in R_{q-1}$ (Lemma 3.2(b)); hence $N_{j-1} \notin R_{q-2}$, i.e. $c_{j-1} \leq q - j - 2$. But $N_j \in R_q$ gives $c_j \geq q - j$, so $c_j \geq c_{j-1} + 2$, contradicting Lemma 3.2(b). $\square$

5 The tail lemma

This is [4, Lemma 3.3(2) and Lemma 4.3(2)], in a form that serves both the triangular and the non-triangular case.

Lemma 5.1. *Let $k \geq 2$ and $t \geq k + 1$ with $c_t = k - 1$, $c_{t+1} = k$, and $c_j \geq k - 1$ for $t \leq j \leq t + k - 2$. Then at least one of the following holds.*

- (A) *There is a ramp (p, q, k) with $t - k \leq p < q \leq t - 1$.*
- (B) *There is a ramp $(p, q, k - 1)$ with $t - k + 1 \leq p < q \leq t + 1$.*
- (C) *Every part of $\lambda^{(t-1)}$ is at most k , and $c_{t+m} = c_{t-k+m}$ for $0 \leq m \leq k - 1$.*

Proof. The k labels $N_{t-k}, \dots, N_{t-1}$ exist ($t-k \geq 1$); as $|R_t| = k-1$, one of them is not in R_t , and $N_{t-1} \in R_t$. Let i be the largest index in $[t-k, t-2]$ with $N_i \notin R_t$. Then $N_{i+1}, \dots, N_{t-1} \in R_t$; as $c_{t+1} = c_t + 1$, $R_t \subseteq R_{t+1}$, so $N_{i+1} \in R_{t+1}$, and Lemma 3.3 gives $c_i = t-i-1$ and $c_{i+1} = t-i$.

Case 1: $i \geq t-k+1$. Then $c_i \leq k-2 < k-1 < k = c_{t+1}$, and Lemma 4.2 gives a ramp $(p, q, k-1)$ with $i \leq p < q \leq t+1$: (B) holds.

Case 2: $i = t-k$. Then $c_{t-k} = k-1$, $c_{t-k+1} = k$, and $R_t = \{N_{t-k+1}, \dots, N_{t-1}\}$ (these $k-1$ labels lie in R_t and $|R_t| = k-1$). If $c_j \leq k-2$ for some $j \in [t-k+2, t-1]$, Lemma 4.2 applied to $j < t+1$ gives (B). If $c_j \geq k+1$ for some $j \in [t-k+2, t-1]$, Lemma 4.2 applied to $t-k < j$ (with $c_{t-k} = k-1 < k < c_j$) gives a ramp (p, q, k) with $t-k \leq p < q \leq j \leq t-1$: (A) holds. Otherwise $c_j \in \{k-1, k\}$ for all $j \in [t-k+1, t-1]$; we show (C). The parts of $\lambda^{(t-1)}$ are the sizes $c_j - (t-1-j) \leq k$ of the piles $N_j \in R_t$. Write $c_{t-1-u} = k-1 + \varepsilon_u$ with $\varepsilon_u \in \{0, 1\}$ ($0 \leq u \leq k-2$). Fix $m \in [0, k-1]$ and determine R_{t+m} :

- labels not in R_t (all O 's and all N_j with $j \leq t-k$) are not in R_{t+m} (Lemma 3.2(d));
- for $t \leq j \leq t+m-1$: $c_j \geq k-1 \geq m \geq t+m-j$, so $N_j \in R_{t+m}$ (m labels);
- $N_{t-1-u} \in R_{t+m} \iff k-1 + \varepsilon_u \geq m+1+u \iff u \leq k-2-m + \varepsilon_u$.

For $m=0$ this gives $c_t = k-1 = c_{t-k}$. For $1 \leq m \leq k-1$ the labels N_{t-1-u} with $u \leq k-2-m$ are in R_{t+m} , the one with $u = k-1-m$ is in iff $\varepsilon_{k-1-m} = 1$, and those with $u \geq k-m$ are not. So $c_{t+m} = m + (k-1-m) + \varepsilon_{k-1-m} = k-1 + \varepsilon_{k-1-m} = c_{t-k+m}$. $\square$

6 Two lemmas for non-triangular n

Lemma 6.1 (too many parts). *Let $n = T_{k-1} + r$ with $1 \leq r \leq k$ and $\lambda \vdash n$. If $c_m \geq k+1$ for some $m \geq 1$, then $d_{\mathcal{B}}(\lambda) \geq m+1$.*

Proof. $\lambda^{(m-1)}$ has $c_m \geq k+1$ parts, so it is not cyclic (Corollary 2.6(b)), and $d_{\mathcal{B}}(\lambda) \geq m$. Suppose $d_{\mathcal{B}}(\lambda) = m$, i.e. $\lambda^{(m)}$ is cyclic. Its iterates are $\lambda^{(m)}, \dots, \lambda^{(m+k-1)}$, whose part counts are $c_{m+1}, \dots, c_{m+k}$, so Corollary 2.6(c) gives $\sum_{i=m+1}^{m+k} c_i = k(k-1) + r$. Lemma 3.5 with $a = m-1$, $h = k$ gives $\sum_{i=m}^{m+k-1} c_i \leq n + T_{k-1} = k(k-1) + r$. Subtracting, $c_m \leq c_{m+k} \leq k$, a contradiction. $\square$

Lemma 6.2 (the anchor t). *Let $k \geq 2$, $n = T_{k-1} + r$ with $1 \leq r \leq k-1$, and $\lambda \vdash n$. The set*

$$\mathcal{T} = \{i \geq 1 : \lambda^{(i-1)} \text{ is cyclic, } c_i = k-1, c_{i+1} = k\}$$

is non-empty; let $t = \min \mathcal{T}$. Then $d_{\mathcal{B}}(\lambda) \leq t-1$, and for all $j \geq t$: $c_j \in \{k-1, k\}$, $c_{j+k} = c_j$, and $\sum_{i=j}^{j+k-1} c_i = k(k-1) + r$.

Proof. Let $d = d_{\mathcal{B}}(\lambda)$. For $i \geq d+1$, $\lambda^{(i-1)}$ is cyclic, so by Corollary 2.6(b),(c) the sequence $(c_i)_{i \geq d+1}$ takes only the values $k-1$ and k , is periodic with period k , and takes both values. Choose $i_0 \geq d+1$ with $c_{i_0} = k-1$, and let $i_1 > i_0$ be least with $c_{i_1} = k$. Then $c_{i_1-1} = k-1$, and $i = i_1 - 1 \geq d+1$ lies in $\mathcal{T}$. If $t \in \mathcal{T}$, then $\lambda^{(t-1)}$ is cyclic, so $d \leq t-1$. The remaining statements hold because every $\lambda^{(j-1)}$ with $j \geq t$ is cyclic, by Corollary 2.6(b),(c). $\square$

7 The upper bound

Theorem 7.1 ([4, Thm. 4.4(1)]). *Let $k \geq 4$ and $n = T_{k-1} + r$ with $1 \leq r \leq k-1$. Then $D_{\mathcal{B}}(n) \leq k^2 - 2k - 1$.*

Proof. $k = 4$ ($n \in \{7, 8, 9\}$, bound 7): by exhaustive computation over the $15 + 22 + 30$ partitions, $D_{\mathcal{B}}(7) = 4$, $D_{\mathcal{B}}(8) = 5$, $D_{\mathcal{B}}(9) = 7$ (Appendix A; this matches [4, Fig. 1]). The general argument below does not cover $k = 4$: in case (B) with $L = k$ it only gives $d \leq T_k - 2 = 8 > 7$.

$k \geq 5$. Let $\lambda \vdash n$, $d = d_{\mathcal{B}}(\lambda)$, and let t be as in Lemma 6.2, so $d \leq t - 1$. If $t \leq k$, then $d \leq k - 1 < k^2 - 2k - 1$. Assume $t \geq k + 1$. By Lemma 6.2 the hypotheses of Lemma 5.1 hold.

(C) is impossible. Suppose $c_{t-k+m} = c_{t+m}$ for $0 \leq m \leq k - 1$. For $m \geq k$, $c_{t-k+m} = c_{t+(m-k)} = c_{t+m}$ by the periodicity in Lemma 6.2. Thus $\lambda^{(t-k-1)}$ and $\lambda^{(t-1)}$ have the same part-count sequence $(c_t, c_{t+1}, \dots)$, so they are equal (Lemma 3.4). In particular $\lambda^{(t-k-1)}$ is cyclic. Since also $c_{t-k} = c_t = k - 1$, $c_{t-k+1} = c_{t+1} = k$ and $t - k \geq 1$, we get $t - k \in \mathcal{T}$, contradicting the minimality of t .

(A): a ramp (p, q, k) with $t - k \leq p < q \leq t - 1$, of length $L = q - p \leq k - 1$. If $L = k - 1$, then $p = t - k$, $q = t - 1$ and $c_{t-1} = k + 1$. Lemma 3.5 ($a = t - 2$, $h = k$) gives $c_{t-1} + \sum_{i=t}^{t+k-2} c_i \leq k(k-1) + r$, while $\sum_{i=t}^{t+k-1} c_i = k(k-1) + r$ by Lemma 6.2. Hence $c_{t-1} \leq c_{t+k-1} \leq k$, a contradiction. If $L \leq k - 2$, Proposition 4.5 gives $p \leq (k - 3)k$, and $d \leq t - 1 \leq p + k - 1 \leq k^2 - 2k - 1$.

(B): a ramp $(p, q, k - 1)$ with $t - k + 1 \leq p < q \leq t + 1$, of length $L \leq k$.

- $L = k$: then $(p, q) = (t - k + 1, t + 1)$, and Lemma 4.6 (with $x = k - 1 \geq 2$) gives $t + 1 \leq n + 1$. So $d \leq t - 1 \leq n - 1 \leq T_{k-1} + k - 2 = T_k - 2$, and $(k^2 - 2k - 1) - (T_k - 2) = \frac{1}{2}(k^2 - 5k + 2) > 0$ for $k \geq 5$.
- $L = k - 1$: then $(p, q) = (t - k + 1, t)$ or $(t - k + 2, t + 1)$. The first would give $c_t = k$, but $c_t = k - 1$. In the second, $c_{t-k+2} = k - 2 = t - (t - k + 2)$, so by (2) $N_{t-k+2} \in R_t \setminus R_{t+1} = Z_t$. But $c_{t+1} = c_t + 1$ forces $Z_t = \emptyset$ (Lemma 3.2(c)). Contradiction.
- $L \leq k - 2$: $p \leq (k - 3)(k - 1)$ by Proposition 4.5, and $d \leq t - 1 \leq p + k - 2 \leq k^2 - 3k + 1 < k^2 - 2k - 1$.

In all possible cases $d \leq k^2 - 2k - 1$. □

8 Equality for $n = T_k - 1$

For $k \geq 3$, $h \in \{1, \dots, k\}$ and $x \in \{1, \dots, k + 1\}$ put

$$X'(h, x) = \Delta_{k-1} \cup (D_k \setminus \{d_k(h), d_k(h \oplus_k 1)\}) \cup \{d_{k+1}(x)\},$$

where for $w \geq 1$ and $1 \leq j \leq w$, $j \oplus_w 1 = j + 1$ if $j < w$ and $w \oplus_w 1 = 1$.

Lemma 8.1. *Let $k \geq 3$ and let μ be a partition with $Y(\mu) = X'(h, x)$. If $(h, x) \neq (k - 1, 1)$, then $Y(\mathcal{B}(\mu)) = X'(h \oplus_k 1, x \oplus_{k+1} 1)$. If $(h, x) = (k - 1, 1)$, then $\mu = (k + 1, k - 1, k - 2, \dots, 3, 1)$ (to be read as $(4, 1)$ when $k = 3$) and $\mathcal{B}(\mu) = (k, k - 1, \dots, 2)$, which is cyclic.*

Proof. Let $H = \{h, h \oplus_k 1\}$, a set of two elements as $k \geq 3$. The cells of row 1 are among the $d_w(w) = (1, w)$ and those of column 1 among the $d_w(1) = (w, 1)$. Counting them, $\ell(\mu) = k - 1 + [k \notin H] + [x = k + 1]$ and $\mu_1 = k - 1 + [1 \notin H] + [x = 1]$. So $\ell(\mu) \leq \mu_1 - 2$ iff $k \in H$, $x \neq k + 1$, $1 \notin H$ and $x = 1$. Now $k \in H$ iff $h \in \{k - 1, k\}$, and $h = k$ gives $H = \{k, 1\} \ni 1$. So this happens iff $(h, x) = (k - 1, 1)$ (note $1 \notin \{k - 1, k\}$ as $k \geq 3$). If $(h, x) \neq (k - 1, 1)$, Corollary 2.4 and Lemma 2.1 give $Y(\mathcal{B}(\mu)) = \rho(X'(h, x)) = X'(h \oplus_k 1, x \oplus_{k+1} 1)$. If $(h, x) = (k - 1, 1)$, the column lengths are $k + 1$ (column 1), $k + 1 - j$ for $2 \leq j \leq k - 2$, 1 (column $k - 1$: the cell of Δ_{k-1} only) and 0 (column k). So $\mu = (k + 1, k - 1, \dots, 3, 1)$ (i.e. $(4, 1)$ for $k = 3$) with $k - 1$ parts, and $\mathcal{B}(\mu)$ has the parts $k - 1$ and $k, k - 2, \dots, 2$ (the part 1 disappears). That is $\mathcal{B}(\mu) = (k, k - 1, \dots, 2) = \lambda_E$ with $E = D_k \setminus \{d_k(k)\}$, which is cyclic by Theorem 2.5. □

Theorem 8.2 ([4, Thm. 4.4(2)]). *For $k \geq 3$, $\lambda^{(k)} = (k-1, k-2, k-2, k-3, \dots, 2, 1, 1) \vdash T_k - 1$ and $d_{\mathcal{B}}(\lambda^{(k)}) = k^2 - 2k - 1$. Consequently $D_{\mathcal{B}}(T_k - 1) = k^2 - 2k - 1$ for all $k \geq 4$.*

Proof. The sequence is non-increasing with sum $(k-1) + (k-2) + T_{k-2} + 1 = T_k - 1$. Its diagram is $X'(1, k+1)$: columns 1 and 2 have $k-1$ and $k-2$ cells (those of Δ_{k-1} , so $d_k(1), d_k(2)$ are missing). Column $j \in [3, k]$ has $k+1-j$ cells (Δ_{k-1} plus $d_k(j)$), and column $k+1$ is $\{d_{k+1}(k+1)\}$. Let $h_m \equiv 1 + m \pmod{k}$ and $x_m \equiv m \pmod{k+1}$ with $h_m \in [1, k]$, $x_m \in [1, k+1]$. Then $(h_m, x_m) = (k-1, 1)$ iff $m \equiv -2 \pmod{k}$ and $m \equiv 1 \pmod{k+1}$. $M = k^2 - 2k - 2$ satisfies both $(k^2 - 2k - 2 \equiv 1 + 2 - 2 \pmod{k+1})$, and for $k \geq 3$, $0 \leq M < k(k+1)$. So M is the least non-negative solution. By Lemma 8.1 and induction, $Y(\mathcal{B}^m(\lambda^{(k)})) = X'(h_m, x_m)$ for $m \leq M$, and $\mathcal{B}^{M+1}(\lambda^{(k)})$ is cyclic. For $m \leq M$ the diagram contains a cell of D_{k+1} , so $\mathcal{B}^m(\lambda^{(k)})$ is not cyclic (Corollary 2.6(b): cyclic diagrams lie in Δ_k). Hence $d_{\mathcal{B}} = M + 1 = k^2 - 2k - 1$. With Theorem 7.1, $D_{\mathcal{B}}(T_k - 1) = k^2 - 2k - 1$ for $k \geq 4$. (For $k = 3$, $D_{\mathcal{B}}(5) = 3 > 2$; for $k = 2$, $D_{\mathcal{B}}(2) = 0$.) $\square$

9 All extremal partitions of $T_k - 1$

Theorem 9.1. *Let $k \geq 5$ and $\lambda \vdash T_k - 1$. The following are equivalent:*

- (1) $d_{\mathcal{B}}(\lambda) = k^2 - 2k - 1$;
- (2) $c_{k^2-2k-2} \geq k+1$, i.e. $\mathcal{B}^{k^2-2k-3}(\lambda)$ has at least $k+1$ parts;
- (3) $(jk, jk+j+1, k)$ is a ramp for every $j = 1, \dots, k-3$.

Proof. (3) $\Rightarrow$ (2): $j = k-3$ gives $c_{k(k-3)+k-2} = c_{k^2-2k-2} = k+1$. (2) $\Rightarrow$ (1): by Lemma 6.1 (here $r = k-1$), $d_{\mathcal{B}}(\lambda) \geq k^2 - 2k - 1$; the reverse inequality is Theorem 7.1. (1) $\Rightarrow$ (3): go through the proof of Theorem 7.1 with $d = k^2 - 2k - 1$. $t \leq k$ would give $d \leq k-1$. (C) and (A) with $L = k-1$ are impossible, and whenever (B) holds it gives $d < k^2 - 2k - 1$ (strictly, as $k \geq 5$). Hence (A) holds with $L \leq k-2$, and $d \leq t-1 \leq p+k-1 \leq (L-1)k+k-1 \leq k^2 - 2k - 1$ is an equality: $L = k-2$ and $p = (L-1)k$. The equality part of Proposition 4.5 gives the ramps $(jk, jk+j+1, k)$ for $j = 1, \dots, k-3$. $\square$

Remark 9.2. As for T_k , this is a dynamical characterisation, not a closed formula. For $k = 5$ the extremal partitions of 14 are $(4, 3, 3, 2, 1, 1) = \lambda^{(5)}$, $(4, 3, 2, 2, 2, 1)$, $(4, 3, 2, 2, 1, 1, 1)$, $(3, 3, 3, 2, 2, 1)$, $(3, 3, 3, 2, 1, 1, 1)$ and $(3, 3, 2, 2, 2, 1, 1)$. For $k = 4$ the unique extremal partition of 9 is $(3, 2, 2, 1, 1)$; the proof of Theorem 9.1 does not apply to $k = 4$, and this fact is from exhaustive computation over the 30 partitions of 9 (Appendix A). Numbers of extremal partitions of $T_k - 1$:

k	4	5	6	7	8	9	10
$\#\{\lambda \vdash T_k - 1 : d_{\mathcal{B}}(\lambda) = k^2 - 2k - 1\}$	1	6	34	175	831	3911	18163

10 Towards a direct description of the extremal partitions

Proposition 10.1 (a proved direct necessary condition). *Let $k \geq 5$ and let $\lambda \vdash T_k - 1$ be extremal. Then*

- (i) $\lambda_1 \leq k+1$;
- (ii) $\lambda'_i := \#\{m : \lambda_m \geq i\} \geq k+3-2i$ for every $i \geq 1$. Equivalently, $Y(\lambda)$ contains the diagram of $\text{lo}_k = (\lceil (k+2-j)/2 \rceil)_{j=1}^{k+1}$, whose conjugate is $(k+1, k-1, k-3, \dots)$;

(iii) $\ell(\lambda) \geq k + 1$ (the case $i = 1$ of (ii)).

Proof. By Theorem 9.1, $(k, k + 2, k)$ is a ramp: $c_k = k - 1$, $c_{k+1} = k$, $c_{k+2} = k + 1$. We claim $R_{k+2} = \{N_1, \dots, N_{k+1}\}$. As $c_{k+1} = c_k + 1$ and $c_{k+2} = c_{k+1} + 1$, Lemma 3.2(c) gives $R_k \subseteq R_{k+1} \subseteq R_{k+2}$, and $N_k, N_{k+1} \in R_{k+2}$. Suppose some N_i with $1 \leq i \leq k - 1$ is not in R_{k+2} , and take i largest. Then $N_{i+1} \in R_{k+2}$, hence $N_{i+1} \in R_{k+1}$ (Lemma 3.2(d)), and $N_i \notin R_k$ (as $R_k \subseteq R_{k+2}$). Lemma 3.3 (with $j = i < k$) gives $N_{i+1} \notin R_{k+2}$, a contradiction. So $N_1, \dots, N_{k+1} \in R_{k+2}$, and since $|R_{k+2}| = c_{k+2} = k + 1$ the claim follows.

(i) No O_m lies in R_{k+2} , i.e. $\lambda_m \leq k + 1$ for all m . (ii) For $1 \leq i \leq k + 1$, $N_i \in R_{k+2}$ gives $c_i \geq k + 2 - i$. Hence for $j < i \leq k + 2$, $c_j \geq k + 2 - j \geq i - j$, i.e. $N_j \in R_i$ by (2). So $|R_i| = c_i$ and (2) give $\lambda'_i = c_i - (i - 1) \geq k + 3 - 2i$. For $i \geq k + 2$ the bound is negative. The conjugate of lo_k is $\#\{j \leq k + 1 : \lceil (k + 2 - j)/2 \rceil \geq i\} = \#\{j \leq k + 3 - 2i\} = k + 3 - 2i$, which gives the equivalent form. $\square$

This is the analogue of condition (1) of [4, Thm. 3.8]. The following observation shows why we do not claim a direct characterisation.

Remark 10.2 (computational observation, not used). Let hi_k be the partition with conjugate $(2k - 3, 2k - 5, \dots, 3, 1)$. For $5 \leq k \leq 10$ every extremal partition of $T_k - 1$ lies between lo_k and hi_k (`bs.py`). But the number of partitions of $T_k - 1$ in this interval is 6, 35, 175, 834, 3935, 18330, against 6, 34, 175, 831, 3911, 18163 extremal ones. So “ $\text{lo}_k \subseteq \lambda \subseteq \text{hi}_k$ ” characterises the extremal partitions for $k = 5, 7$ but not for $k = 6, 8, 9, 10$. Griggs–Ho made the same kind of observation in the triangular case: their three direct conditions [4, Thm. 3.8] are necessary but not sufficient ($k = 8$).

What Theorem 9.1 does and does not give. It is an exact “if and only if” description of the set of extremal partitions of $T_k - 1$, and it goes beyond the bare definition in two ways. First, (2) is a test on a single number: the part count at one fixed time $k^2 - 2k - 2$ must be at least $k + 1$. The definition, by contrast, asks whether $\mathcal{B}^{k^2 - 2k - 3}(\lambda)$ is cyclic, which needs Brandt’s list. Second, (3) says that every extremal orbit has the same rigid skeleton: ramps of value k at the fixed positions $jk, \dots, jk + j + 1$, $j = 1, \dots, k - 3$. However, it is a *dynamical* description, not a closed-form (static) description of the diagrams. Proposition 10.1 extracts direct conditions from the first ramp only. Remark 10.2 shows that the natural static bounds do not suffice. We do not have a proof that no simple static description exists; the heuristic obstacle is that the ramps with $j \geq 2$ involve the orbit up to time about k^2 . We therefore regard the question “which partitions attain the maximum” as answered exactly but not in closed form, and we leave a closed-form description open.

11 Computations

Computations used in proofs: the case $k = 4$ of Theorem 7.1 (exhaustive over the 67 partitions of 7, 8, 9) and the uniqueness of the extremal partition of 9 (exhaustive over its 30 partitions). Both are done by the script in Appendix A, which runs in well under a second. As sanity checks, `bs.py` (exhaustive for $n \leq 60$, about 1.5 minutes) confirms: $D_{\mathcal{B}}(n) \leq k^2 - 2k - 1$ for all non-triangular $n \leq 60$ with $k \geq 4$; $D_{\mathcal{B}}(T_k - 1) = k^2 - 2k - 1$ for $4 \leq k \leq 10$; $d_{\mathcal{B}}(\lambda^{(k)}) = k^2 - 2k - 1$ for $3 \leq k < 80$; and Theorem 9.1 for $5 \leq k \leq 10$. It also checks the ramp bound and Lemma 4.6 on every ramp among the first $4n + 10$ part counts of every partition of $n \leq 30$.

A Code for the exhaustive small cases

File `p3_small_cases.py` (Python 3, no dependencies; runs in well under one second). It computes $d_{\mathcal{B}}$ directly from the definition: the orbit of λ is followed until a partition repeats, and the index of the first occurrence of that partition is $d_{\mathcal{B}}(\lambda)$.

```
#!/usr/bin/env python3
"""Exhaustive small cases used in p3_c2/p3_c3/p3_c4/p3_c5.
Straight from the definitions: B(lambda), d_B(lambda) = number of steps before the
first partition that recurs in the orbit, D_B(n) = max over all partitions of n.
Runtime: well under one second."""

def partitions(n, maxp=None):
    maxp = n if maxp is None else maxp
    if n == 0:
        yield ()
        return
    for p in range(min(n, maxp), 0, -1):
        for rest in partitions(n - p, p):
            yield (p,) + rest

def B(lam):
    return tuple(sorted([len(lam)] + [x - 1 for x in lam if x > 1], reverse=True))

def d_B(lam):
    first = {}
    i = 0
    while lam not in first:
        first[lam] = i
        lam = B(lam)
        i += 1
    return first[lam] # the tail length: B^i(lam) is cyclic iff i >= this

def fmt(lam):
    # compact notation: (3,3,2,1) -> 3^2 2 1
    out = []
    for v in sorted(set(lam), reverse=True):
        m = lam.count(v)
        out.append(f"{v}^{m}" if m > 1 else f"{v}")
    return "(" + " ".join(out) + ")"

for n in [1, 3, 4, 5, 6, 7, 8, 9, 12, 17]:
    parts = list(partitions(n))
    d = {lam: d_B(lam) for lam in parts}
    D = max(d.values())
    ext = [lam for lam in parts if d[lam] == D]
    print(f"n={n}: {len(parts)} partitions, D_B={D}, extremal: {'', ' '.join(fmt(l) for l in ext)}")
```

Output:

```
n=1: 1 partitions, D_B=0, extremal: (1)
n=3: 3 partitions, D_B=2, extremal: (1^3)
n=4: 5 partitions, D_B=2, extremal: (1^4)
n=5: 7 partitions, D_B=3, extremal: (1^5)
n=6: 11 partitions, D_B=6, extremal: (2^2 1^2)
```

n=7: 15 partitions, D_B=4, extremal: (1^7)
 n=8: 22 partitions, D_B=5, extremal: (1^8)
 n=9: 30 partitions, D_B=7, extremal: $(3\ 2^2\ 1^2)$
 n=12: 77 partitions, D_B=8, extremal: $(3^2\ 2^2\ 1^2)$, (1^{12})
 n=17: 297 partitions, D_B=12, extremal: (1^{17})

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